

Mohammadreza Qaderi

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SUMMARY

Senior Backend Engineer with 5+ years of experience designing high-performance, distributed systems using Go and Python. Proven expertise in microservices architecture, financial trading platforms, and engineering scalable data pipelines capable of processing millions of daily events. Adept at leading technical teams and optimizing cloud-native infrastructure.

EDUCATION

- Tarbiat Modares University** Tehran, Iran
M.Sc. in Computer Engineering – Computer Networks; GPA: 4.0 Sept 2024 - Present
- Amirkabir University of Technology** Tehran, Iran
B.Sc. in Computer Software Engineering; GPA: 3.89 Sept 2018 - Jun 2023
Thesis: A Paper Trading System for Iranian Stocks
- National Organization for Development of Exceptional Talents (SAMPAD)** Tehran, Iran
Diploma in Mathematics and Physics; GPA: 4.0 Sept 2014 - Aug 2018

WORKING EXPERIENCE & RESEARCH

- Backend Tech Lead & Software Engineer** Pishro Broker
Aug 2023 - Present
 - Microservices Architecture Design:** Architected and transitioned legacy monoliths into fully isolated, gRPC-based microservices, integrating independent databases, RabbitMQ for message queuing, and Redis for caching to enhance system scalability.
 - High-Performance Data Pipelines:** Engineered distributed data scraping schedulers to ingest and process 10,000 to 50,000 daily market records per node, alongside post-market computational schedulers for financial data aggregation.
 - Infrastructure & Deployment:** Setup CI/CD pipeline using GitLab to lint and build services. Containerized the entire microservice ecosystem using Docker and Docker Compose, and utilized MinIO for highly scalable and reliable object storage.
 - Core Platform Development:** Led backend development for critical trading platforms and infrastructure gateways, significantly impacting my.pishrobroker.ir, gateway.pishrobroker.ir, and optionz.ir.
- Chief Technology Officer (CTO) & Team Lead** SefroYek Farda
Jan 2021 - Present
 - Platform Architecture & Scaling:** Led the backend system architecture for B2B and B2C educational platforms (Entekhab Baazmoon, Hedayat Baazmoon), successfully scaling the infrastructure to support over 80,000 active users.
 - AI Integration & Automation:** Spearheaded the integration of Artificial Intelligence for personalized academic recommendations and engineered an automated reporting pipeline generating custom Microsoft Word documents for users.
- Software Engineer (Part-Time)** Farasima
May 2024 - Present
 - High-Throughput SMS Firewall:** Contributed to the backend development of a high-performance SMS firewall system capable of processing and rule-matching 300,000 to 500,000 messages per second.
 - gRPC API & Gateway Integration:** Designed gRPC-based communication interfaces for external gateways to enforce firewall regulations and seamlessly route specific messages to AI-based anomaly detection modules.
 - System Monitoring & Observability:** Established comprehensive system monitoring, metrics collection, and alerting pipelines using Prometheus and Grafana to ensure high availability of critical nodes.
 - Data Infrastructure:** Leveraged ClickHouse for ultra-fast analytical data storage and Apache Superset for internal data visualization, routed securely through HAProxy.
- Software Engineer** GradientDp!
Mar 2022 - Dec 2022
 - Paper Trading & Portfolio Management:** Designed and maintained a paper trading system based on the Iranian stock market, enabling users to simulate trading with real-time market data and analyze their strategic performance metrics.
- AI Trainee at Advanced Technologies Development Research** Sharif University of Technology
Aug 2019 - Sept 2020
 - Video Swin Transformer:** Researched and implemented Video Swin Transformer models to improve speed-accuracy trade-offs in video recognition tasks by leveraging an inductive bias of locality and spatial-temporal factorization.
- Software Engineer** Snapp!
Mar 2019 - Aug 2019
 - Event Dispatching:** Designed and developed event dispatching services based on polling and MQTT as a core Dispatching team member.
 - Infrastructure Management:** Deployed and maintained high-traffic, heavy-load services on Kubernetes clusters.

TEACHING EXPERIENCE

- **Head Teaching Assistant & Lab Instructor** Amirkabir University of Technology
Fall 2020 - Spring 2026
Faculty of Computer Engineering
 - **Head Teaching Assistant:** Computer Networks, Startup Development, Data Mining, Web Engineering, Operating Systems.
 - **Teaching Assistant:** Internet of Things, Microprocessor and Assembly, Data Structures and Algorithms, Design Algorithms, Advanced Programming.
 - **Lab Instructor:** Microprocessor and Assembly Labs (Instructed across 6 consecutive semesters).

NOTABLE PROJECTS

- **University Major Selection System:** Developed an AI-driven platform for Iranian university applicants to analyze entrance exam results and predict acceptance probabilities based on historical statistics and personal interests.
- **Paper Trading Platform:** Built a simulated trading environment for the Iranian stock market, allowing users to safely backtest strategies and understand market dynamics with real-time data.
- **Video Swin Transformer:** Implemented a machine vision model based on Video Swin Transformers for advanced video behavior recognition and analysis.
- **Search Engine:** Engineered a document retrieval system leveraging both statistical models and word embeddings to efficiently process and answer user queries.
- **Evolutionary Game AI (Flappy Plane):** Designed an evolutionary algorithm capable of learning and mastering a game across 3 different difficulty modes over multiple generations.
- **Handwritten Digit Recognition:** Implemented a fully connected Artificial Neural Network (ANN) from scratch using NumPy to classify the MNIST dataset.
- **Super Mario Bros in Java:** Developed a Super Mario-style 2D platformer desktop application in Java, featuring custom game state management, multiplayer mechanics, and social music sharing.
- **Message Broker:** Designed a custom message broker and robust text messenger using Python's Socket library, enabling reliable multi-client communication.
 - *Explore more projects on GitHub or my Personal Homepage.*

SKILLS & QUALITIES

- **Programming Languages:** Python 3, Go, C, C++, JavaScript, Java, PHP, $\LaTeX$.
- **Tools & DevOps:** Docker, Kubernetes, PM2, Airflow, Nginx, HAProxy, Git, Linux, Bash.
- **Frameworks & Libraries:** GoFiber, Echo, Django, FastAPI, NumPy, Pandas, React.js, D3.js. (Working knowledge: React Native, Flutter)
- **Databases:** PostgreSQL, MongoDB, SQL Server, Elasticsearch, ClickHouse.
- **Languages:** Persian (Native), Turkish (Native), English (Proficient - TOEFL-iBT: 108/120).

REFERENCES

- **Dr. Hamed Farbeh**
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